Exercise 8:

Following the method of the previous slide and considering the backward Taylor expansion shown in eq. (20), write the first order approximation and truncation error of

$$U(x - \Delta x) =$$

Rearrange the equation to obtain the velocity gradient

$$\frac{\partial U}{\partial x} =$$

and show the **backward in space numerical approximation** (sometimes called upstream) using the discrete indices (define them accordingly)

$$\frac{\partial U}{\partial x} \approx$$

Upload either a picture of your workings or a document.

$$U(x - \Delta x) = U(x) - \frac{\partial U}{\partial x}\Delta x + O(\Delta x^2)$$

$$\frac{U(x - \Delta x) - U(x) - O(\Delta x^2)}{-\Delta x} = \frac{\partial U}{\partial x}$$

velocity gradient $$\frac{\partial U}{\partial x} = -\frac{[U(x - \Delta x) - U(x)]}{\Delta x} + O(\Delta x)$$

backward approximation $$\frac{\partial U}{\partial x} \approx -\frac{[U^n_{i+1} - U^n_i]}{\Delta x}$$

Exercise 9:

- Write the grid-based, discretized form of eq. (29), first defining the discrete time and space axes to be used as indices

$$t^n = \quad ; x_i =$$

$$\frac{\partial^2 U}{\partial x^2} \approx$$

- Given the discrete 3-D horizontal velocity field U^n_{ijk}, write the discretized form of eq. (33)
- As usual, upload a picture of your paper or a document

$$t^n = t_0 + n(\Delta t)$$
$$x_i = x_0 + i(\Delta x)$$

Equation 29: $$\frac{\partial^2 U}{\partial x^2} = \frac{U(x + \Delta x) - 2U(x) + U(x_i - \Delta x)}{\Delta x^2} + O(\Delta x^2)$$

$$\frac{\partial^2 U}{\partial x^2} = \frac{U(x_{i+1}) - 2U(x_i) + U(x_{i-1})}{\Delta x^2} + O(\Delta x^2)$$

$$\frac{\partial^2 U}{\partial x^2} \approx \frac{U^n_{i+1} - 2U_i + U^n_{i+1}}{\Delta x^2} + O(\Delta x^2)$$

Equation 33: $$\frac{\partial U}{\partial z} = \frac{U(z_i + \Delta z) - U(z_i - \Delta z)}{2\Delta z} + O(\Delta z^2)$$

$$\frac{\partial U}{\partial z} \approx \frac{U^n_{ij(k+1)} - U^n_{ij(k-1)}}{2\Delta z} + O(\Delta z^2)$$

Exercise 4:

- Calculate the Taylor series expansion for the following univariate functions $f(x)$ about the point $x = x_0$. Only show the series terms until the truncation error is $\mathcal{O}\left((x-x_0)^3\right)$ and upload a picture or a document on your GitHub:
  1. $f(x) = \frac{x}{x+1}$
  2. $f(x) = \frac{2x^2}{x^4}$
  3. $f(x) = \ln x^2$

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + O((x-x_0)^3)$$

1. $f(x_0) = \frac{x_0}{x_0+1}$ $\quad f'(x) = \frac{1}{(x+1)^2}$ $\quad f''(x) = \frac{-2}{(x+1)^3}$

$f'(x_0) = \frac{1}{(x_0+1)^2}$ $\quad f''(x_0) = \frac{-2}{(x_0+1)^3}$

$$f(x) = \frac{x_0}{x_0+1} + \frac{x-x_0}{(x_0+1)^2} - \frac{(x-x_0)^2}{(x_0+1)^3} + O((x-x_0)^3)$$

2. $f(x_0) = \frac{2}{x_0^2}$ $\quad f'(x) = -4/x^3$ $\quad f''(x) = 12/x^4$

$f'(x_0) = -4/x_0^3$ $\quad f''(x_0) = 12/x_0^4$

$$f(x) = \frac{2}{x_0^2} - \frac{4(x-x_0)}{x_0^3} + \frac{6(x-x_0)^2}{x_0^4} + O((x-x_0)^3)$$

3. $f(x_0) = 2\ln x_0$ $\quad f'(x) = 2/x$ $\quad f''(x) = -2/x^2$

$f'(x_0) = 2/x_0$ $\quad f''(x_0) = -2/x_0^2$

$$f(x) = 2\ln x_0 + \frac{2(x-x_0)}{x_0} - \frac{(x-x_0)^2}{x_0^2} + O((x-x_0)^3)$$

Exercise 5:

- Consider the implicit *Euler backward* scheme

$$C^{n+1} = \frac{C^n}{(1+k\Delta t)}$$

- This scheme is unconditionally stable
- Demonstrate that this is true by using the same method used in the previous slide. Upload a picture or a document

$C^{n+1} = \frac{C^n}{1+k\Delta t} = \frac{1}{1+k\Delta t} \cdot C^n$

amplification factor: $\lambda = \frac{1}{1+k\Delta t}$

for stability: $|\lambda| \leqslant 1$

$|\lambda| = \left|\frac{1}{1+k\Delta t}\right| = \frac{1}{1+k\Delta t}$

for $k > 0$ and $t > 0$: $1 + k\Delta t > 1$ so $\frac{1}{1+k\Delta t} < 1$

this is true for all positive values of k and Δt so $|\lambda| < 1$ for all positive values of k and Δt